A non-adapted subcase of the antisymmetric weak-subordination problem

arXiv automatic math run `fa_banach_001`

2026-06-18

Source and open target

The source paper is Yaroslavtsev, *Fourier multipliers and weak differential subordination of martingales in UMD Banach spaces*, arXiv:1703.07817v5 [1]. In Section 5.2 it records Conjecture 5.7, a linear-in-$\beta_{p,X}$ weak differential subordination estimate for continuous martingales. The paper then proves Theorem 5.10 for stochastic integrals $(\Phi A) \cdot W^H$ when A is self-adjoint, and states the following open antisymmetric variant.

> The last inequality holds because of structure of Φ so that one can rewrite $(\Phi h_n) \cdot W^H(h_n)$ as a summation in time, and because $(W^H(h_n))_{n\geq 1}$ is a sequence of independent Wiener processes. □
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> **Remark 5.11.** Theorem 5.10 in fact can be shown using [21, Proposition 3.7.(i)].
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> **Remark 5.12.** An analogue of Theorem 5.10 for antisymmetric A (i.e. A such that $A^* = -A$) remains open. It is important for instance for the possible estimate (5.2). Indeed, in Proposition 5.2 the Hilbert space H can be taken 2-dimensional, $A = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, and $\Phi : \mathbb{R}_+ \times \Omega \to \mathcal{L}(H, X)$ is such that $\Phi\left(\begin{smallmatrix} a \\ b \end{smallmatrix}\right) = af_1 + bf_2$ for each $a, b \in \mathbb{R}$. Then $M = \Phi \cdot W^H$, $N = (\Phi A) \cdot W^H$, and if one shows (5.11) for an antisymmetric operator A, then one automatically gains (5.2).

Thus the open point is the adapted-integrand estimate

$$\|((\Phi A) \cdot W^H)_\infty\|_{L^p(\Omega;X)} \leq C_p \beta_{p,X} \|A\| \|(\Phi \cdot W^H)_\infty\|_{L^p(\Omega;X)}$$

for antisymmetric A, especially the two-dimensional rotation which yields the Hilbert-transform estimate.

Result

The following partial result shows that the obstruction in Remark 5.12 is not the antisymmetry of A by itself. It is the interaction with Brownian-path adaptivity of the integrand.

Theorem 1. *Let H be a real Hilbert space, X a finite-dimensional real Banach space, $1 < p < \infty$, and W^H an H-cylindrical Brownian motion. Let $\mathcal{G}$ be a sigma-field independent of W^H. Let $\Phi : [0, T] \times \Omega \to \mathcal{L}(H, X)$ be elementary and $\mathcal{B}([0, T]) \otimes \mathcal{G}$-measurable. Then for every $A \in \mathcal{L}(H)$,*

$$\left(\mathbb{E}\|((\Phi A) \cdot W^H)_T\|^p\right)^{1/p} \leq \|A\| \left(\mathbb{E}\|(\Phi \cdot W^H)_T\|^p\right)^{1/p}.$$

Consequently, for deterministic or initially random Brownian-independent integrands, the antisymmetric case in Remark 5.12 holds with constant 1 for every contraction A.

Proof Intuition

For deterministic Φ, both $(\Phi A) \cdot W^H$ and $\Phi \cdot W^H$ are centered Gaussian vectors in the finite-dimensional Banach space X. If $\|A\| \leq 1$, the covariance form of the first is dominated by the covariance form of the second. The missing covariance is itself a positive covariance form, so one may add an independent Gaussian remainder R to $(\Phi A) \cdot W^H$ and recover the law of $\Phi \cdot W^H$. Jensen's inequality for the convex function $x \mapsto \|x\|^p$ then gives the estimate. Conditioning on the initial randomness gives the same argument for $\mathcal{G}$-measurable Φ.

The source problem remains hard precisely because a genuinely adapted Φ destroys this terminal Gaussian comparison.

Proof

The case $A = 0$ is trivial. By scaling A, it is enough to treat $\|A\| \leq 1$. First suppose that Φ is deterministic and elementary. Set

$$M = (\Phi \cdot W^H)_T, \qquad N = ((\Phi A) \cdot W^H)_T.$$

Both are centered Gaussian X-valued random variables. For $x^* \in X^*$, their covariance forms are

$$q_M(x^*) = \mathbb{E}|\langle M, x^* \rangle|^2 = \int_0^T \|\Phi(t)^* x^*\|_H^2 \, dt$$

and

$$q_N(x^*) = \mathbb{E}|\langle N, x^* \rangle|^2 = \int_0^T \|A^* \Phi(t)^* x^*\|_H^2 \, dt.$$

Since $\|A\| \leq 1$,

$$q_N(x^*) \leq q_M(x^*) \qquad (x^* \in X^*).$$

Thus $q_R := q_M - q_N$ is a positive semidefinite quadratic form on X^*. Because X is finite-dimensional, there exists a centered Gaussian X-valued random variable R, independent of N, whose covariance form is q_R. The centered Gaussian variables $N + R$ and M have the same covariance form, hence the same law. Therefore, using Jensen's inequality conditionally on N,

$$\mathbb{E}\|N\|^p = \mathbb{E}\big\|\mathbb{E}(N + R \mid N)\big\|^p \leq \mathbb{E}\big(\mathbb{E}(\|N + R\|^p \mid N)\big) = \mathbb{E}\|N + R\|^p = \mathbb{E}\|M\|^p.$$

This proves the assertion for deterministic elementary Φ and $\|A\| \leq 1$. Scaling gives the displayed estimate for arbitrary bounded A.

Now let Φ be $\mathcal{B}([0,T]) \otimes \mathcal{G}$-measurable with $\mathcal{G}$ independent of W^H. Conditioning on $\mathcal{G}$ freezes Φ as a deterministic elementary integrand, so the preceding estimate holds conditionally. Integrating the conditional inequality gives the theorem.

Verification notes and limitations

The proof applies to the non-adapted, or initially randomized but Brownian-independent, subcase. It does not prove Conjecture 5.7 and does not prove the adapted antisymmetric analogue of Theorem 5.10 from the source paper. In the adapted case, Φ can depend on the Brownian path driving the integral; the terminal integrals are no longer simply comparable centered Gaussian vectors after conditioning, and the independent-remainder argument has no direct extension.

The result is nevertheless a clean partial answer to Remark 5.12: for deterministic integrands the estimate holds not only for antisymmetric A, but for every contraction A, with constant 1 rather than $\beta_{p,X}$.

Local cheap-index search found no prior packet or attempt for arXiv:1703.07817. A bounded web search on June 18, 2026 for arXiv:1703.07817, weak differential subordination, the anti-symmetric case, and the Hilbert-transform linear estimate found the source paper, the follow-up martingale-decomposition paper arXiv:1706.01731, and related strongly orthogonal martingale work arXiv:1812.08049, but no existing run record for this deterministic subcase.

References

[1] I. Yaroslavtsev. *Fourier multipliers and weak differential subordination of martingales in UMD Banach spaces.* arXiv:1703.07817v5, 2017.